

**Notes: Kacperczyk, Van Nieuwerburgh, and Veldkamp
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Time-Varying Fund Manager Skill

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1 Big picture

- **Question.** Do active mutual fund managers have skill, and if so, what kind of skill do they use over the business cycle? Is the same manager good at stock picking and market timing, or are these separate talents?
- **Setting.** The paper studies actively managed U.S. domestic equity mutual funds from 1980 to 2005 using CRSP mutual fund data, Thomson Reuters holdings, CRSP stock returns, and manager-level information.
- **Core challenge.** Earlier work often finds some evidence of stock-picking ability but little evidence of market timing. A central concern is that these skills are usually measured *unconditionally*, even though the value of processing firm-specific versus aggregate information should vary with the state of the economy.
- **Main conceptual contribution.** The paper redefines managerial skill as *general cognitive ability* to process information and allocate attention across tasks. A skilled manager can pick stocks in booms and time the market in recessions, because the optimal task mix changes with macroeconomic conditions.
- **Main hypotheses.**
 - Managers should display stronger *stock picking* in expansions and stronger *market timing* in recessions.
 - The best managers should vary their use of skill the most over the cycle.
 - The same funds that pick stocks well in booms should also time the market well in recessions.
 - A real-time skill measure that weights timing more in recessions and picking more in booms should predict future fund performance better than either component alone.
- **Main empirical design.** The authors build fund-level measures of timing and picking directly from holdings. They then regress those skill measures on recession indicators, examine heterogeneity across the cross-section of funds, test whether the same funds switch strategies, and construct a real-time *Skill Index* using a recession probability weight.
- **Main baseline result.** Average market-timing ability rises in recessions by about 14 basis points per month, while stock-picking ability falls by about 14 basis points per month. The pattern is economically meaningful and robust to different recession indicators.
- **Main cross-sectional result.** The cyclical change in skill is much stronger in the right tail of the skill distribution. Median managers change less; the best managers change the most.
- **Main “same-manager” result.** Funds that are top stock pickers in expansions are also significantly better market timers in recessions. Those strategy switchers earn roughly 48 to 82 basis points more alpha per year than other funds, depending on the benchmark model.
- **Main real-time result.** A one-standard-deviation increase in the real-time Skill Index predicts about 2.4% higher annualized CAPM alpha one month ahead and about 1.0% to 1.2% higher annualized multifactor alpha; similar predictive power remains at the one-year horizon.
- **Economic takeaway.** The paper argues that mutual fund skill is not a fixed, one-dimensional trait. The strongest managers flexibly shift from micro information processing in booms to macro information processing in recessions.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper studies a large universe of actively managed U.S. domestic equity mutual funds, precisely the setting where the stock-picking versus market-timing debate matters most.
- The economic importance is clear: U.S. households delegated a large and growing share of equity investment to intermediaries, and most domestic equity mutual fund assets were actively managed.
- Because the data include both returns and portfolio holdings, the paper can measure *how* managers create value rather than just whether they outperform.

Interpretation. This is not just a returns paper. The portfolio holdings let the authors distinguish whether good outcomes come from stock selection or from aggregate market exposure.

2.2 Main data sources

- **CRSP Survivorship-Bias-Free Mutual Fund Database:** fund returns and characteristics such as size, age, expenses, turnover, and load.
- **Thomson Reuters holdings data:** reported portfolio holdings, usually semiannual but often quarterly.
- **CRSP stock data and CRSP/Compustat stock-level data:** stock returns, market capitalizations, book-to-market ratios, and momentum.
- **Manager-level sources:** CRSP, Morningstar, Nelson's Directory of Investment Managers, ZoomInfo, and Zabasearch.

2.3 Sample construction

- The sample covers **January 1980 to December 2005**, for a total of 312 months.
- The authors focus on **domestic open-end diversified equity funds** and exclude index funds, sector funds, bond funds, balanced funds, and portfolios with less than 80% in common stock.
- Different share classes are aggregated to the fund level because they share the same underlying portfolio.
- To reduce incubation bias, the authors remove suspicious early observations and also exclude very small funds: those with less than \$5 million in assets or fewer than 10 stocks.
- The final merged sample contains **3,477 distinct funds** and **250,219 fund-month observations**. The manager mapping yields **4,267 managers**.

Interpretation. The empirical variation comes from a very broad panel of funds over many business cycle states, not from a handful of famous managers.

2.4 Business-cycle indicators

- The baseline recession indicator is the **NBER recession dummy**. Out of the 312 months, **38 months** are recession months.
- The paper also uses two more timely macro indicators:
 - **RecRT**: Chauvet–Piger real-time recession probability.
 - **RecCFNAI**: minus the Chicago Fed National Activity Index, so larger values correspond to weaker macro conditions.

Interpretation. The NBER measure is best for ex post evaluation of whether the manager used the right strategy at the right time. The real-time indicators address the concern that managers do not know the NBER dating in real time.

2.5 Defining timing and picking

The paper builds two holdings-based skill measures.

For fund j at time t , **market timing** is

$$\text{Timing}_t^j = \sum_{i=1}^{N^j} \left(w_{i,t}^j - w_{i,t}^m \right) \left(\beta_{i,t} R_{t+1}^m \right), \quad (1)$$

where:

- $w_{i,t}^j$ is fund j 's portfolio weight in stock i at the start of month t ,
- $w_{i,t}^m$ is the market weight of stock i ,
- $\beta_{i,t}$ is the stock's market beta estimated over a 12-month rolling window,
- R_{t+1}^m is the realized market return between t and $t + 1$.

For the same fund, **stock picking** is

$$\text{Picking}_t^j = \sum_{i=1}^{N^j} \left(w_{i,t}^j - w_{i,t}^m \right) \left(R_{i,t+1} - \beta_{i,t} R_{t+1}^m \right). \quad (2)$$

Interpretation.

- Timing asks whether the manager tilts toward high-beta names before aggregate returns are high and away from them before aggregate returns are low.
- Picking asks whether the manager overweights stocks that subsequently earn high *idiosyncratic* returns.
- Both measures are hypothetical return measures based on beginning-of-period holdings.

2.6 Control variables

The baseline regressions include the following fund-level controls:

- $\log(\text{Age})$,

- $\log(\text{TNA})$ for total net assets,
- expense ratio,
- turnover,
- fund flow,
- load,
- style controls along the **size**, **value**, and **momentum** dimensions.

Flow and turnover are winsorized at the 1% level. The style controls summarize the characteristics of the stocks held by the fund.

2.7 Auxiliary objects used later

- **Top fund indicator.** A fund is labeled *Top* if it lies in the top 25% based on how frequently it appears in the upper quartile of Picking during expansions.
- **Alphas.** The paper uses rolling-window CAPM, Fama–French three-factor, and Carhart four-factor alphas, mainly based on returns before expenses when evaluating manager skill and net returns when taking the investor perspective.
- **Portfolio dispersion.** To study herding, the paper defines

$$\text{Portfolio Dispersion}_t^j = \sum_{i=1}^{N^j} \left(w_{i,t}^j - w_{i,t}^m \right)^2. \quad (3)$$

2.8 Descriptive economic facts

- The average monthly entry for timing and picking around zero in expansions is by construction due to demeaning and the interpretation of the constant term.
- The paper emphasizes that the average actively managed fund need not outperform. The relevant question is whether a *subset* of managers has skill and varies how it uses that skill over the cycle.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The core idea is that skill is not a fixed “stock-picking” or “market-timing” trait. Instead, skill is the general ability to acquire and process information. The optimal use of that ability depends on the business cycle:

- in **expansions**, firm-specific information is relatively more valuable, so skilled managers pick stocks;
- in **recessions**, aggregate risk and the price of macro information are more important, so skilled managers time the market.

Interpretation. The paper is really about *task allocation*. Good managers are not permanently “pickers” or permanently “timers.” They reallocate attention as conditions change.

3.2 Baseline panel regressions

The main test regresses the two skill measures on recession states:

$$\text{Picking}_t^j = a_0 + a_1 \text{Recession}_t + a_2 X_t^j + u_t^j, \quad (4)$$

$$\text{Timing}_t^j = b_0 + b_1 \text{Recession}_t + b_2 X_t^j + \varepsilon_t^j. \quad (5)$$

Because the controls are demeaned:

- a_0 is average picking skill in expansions,
- a_1 is the change in picking skill in recessions,
- b_0 is average timing skill in expansions,
- b_1 is the change in timing skill in recessions.

Expected signs.

- $b_1 > 0$: timing improves in recessions.
- $a_1 < 0$: stock picking deteriorates in recessions relative to expansions.

3.3 Inference and panel structure

- The authors estimate pooled panel regressions.
- Standard errors are clustered by **fund** and **time** because recession is common across all funds in a given month.
- They also verify robustness with alternative clustering schemes, including fund-only clustering, style-level clustering, and fund-family clustering.

Interpretation. Since the recession regressor varies only by time, getting the time dimension of uncertainty right is important.

3.4 Quantile regressions

To ask whether only a subset of managers behaves this way, the paper re-estimates the baseline models at the 50th, 75th, and 95th percentiles of the cross-sectional distribution of Timing and Picking.

Interpretation. If cyclical task switching is a feature of skilled managers, the recession coefficient should be bigger in the right tail than at the median.

3.5 Top-fund switching regression

After defining the top-quartile expansion stock pickers, the paper estimates, separately in recessions and expansions,

$$\text{Ability}_t^j = c_0 + c_1 \text{Top}_t^j + c_2 X_t^j + e_t^j, \quad (6)$$

where Ability is either Timing or Picking.

Interpretation.

- In expansions, Top should load positively on Picking.
- In recessions, the same Top funds should load positively on Timing.
- If the same funds are good at both tasks in the appropriate state, that supports the “general skill” interpretation.

3.6 Manager-level regressions

The paper repeats the cyclical skill regressions at the **manager** level and also adds manager fixed effects.

Interpretation. This separates skill embodied in managers from skill embodied purely in fund organizations.

3.7 How timing is implemented

The paper studies the actual portfolio margin through which timing shows up:

- higher cash positions in recessions,
- lower portfolio equity beta in recessions,
- sector rotation toward defensive sectors in recessions and away from cyclical sectors.

Interpretation. Timing is not a black-box regression coefficient. The authors show how managers operationally change portfolios when macro conditions worsen.

3.8 Real-time Skill Index

The real-time manager skill signal is:

$$\text{Skill Index}_{t+1}^j = w_t \text{Timing}_t^j + (1 - w_t) \text{Picking}_t^j, \quad (7)$$

where w_t is the **real-time recession probability**. Timing and Picking are each normalized cross-sectionally to have mean zero and standard deviation one each period.

Interpretation.

- When recession probability is high, the index weights timing more.
- When recession probability is low, it weights stock picking more.
- The index is designed so an investor could compute it in real time at the end of month $t + 1$ without using future information.

3.9 Return forecasting exercises

The paper evaluates the Skill Index in two ways:

- **Portfolio sorts:** each month, funds are sorted into skill quintiles and subsequent abnormal returns and alphas are tracked.

- **Panel predictive regressions:** future one-month-ahead or one-year-ahead alphas are regressed on the current Skill Index and standard fund characteristics.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main result

There is **no external instrument** in the baseline analysis. Identification comes from comparing the measured timing and picking performance of funds across business-cycle states, conditional on fund characteristics. The paper’s credibility therefore depends on whether the cyclical pattern is really about how managers use skill, rather than about sample composition or mechanical return properties.

4.2 Why the recession result is not just an artifact of ex post dating

The baseline regressions use NBER recessions because they are the cleanest ex post classification of economic states. But the paper also re-estimates the models using **real-time recession probabilities** and the **CFNAI**. The same qualitative pattern survives.

Interpretation. The result is not driven by using a recession chronology that managers could not have observed in real time.

4.3 Common unobservables and fixed effects

The paper shows that the cyclical result survives:

- **fund fixed effects**, which rule out changes in the set of active funds;
- **manager fixed effects**, which rule out changes in the set of active managers.

Interpretation. The key pattern is not just that “different kinds of managers show up in recessions.” The same funds and the same managers appear to change how they create value.

4.4 Composition effects

The paper explicitly considers three composition stories:

- **fund-level composition:** the active fund sample changes over the cycle;
- **size-driven composition:** certain strategies simply manage more assets in recessions or booms;
- **manager-level composition:** entry, exit, demotion, or promotion of managers changes the observed averages.

All three are rejected or rendered implausible by the persistence of results under fixed effects and by the evidence that the same top funds switch strategies.

4.5 Mechanical stock-return patterns

A strong alternative is that time variation in stock returns mechanically generates time variation in Timing and Picking even if fund strategies are fixed. The authors simulate stock returns under CAPM-style data-generating processes and feed them into mechanical portfolio rules. None of these artificial strategies generates the observed pattern of better timing in recessions and better picking in booms.

Interpretation. The main result is not a trivial artifact of how the skill measures are constructed.

4.6 Volatility and dispersion effects

Another alternative is that the result reflects high-volatility periods or high earnings-dispersion periods, rather than recessions per se. The paper adds indicators for high aggregate volatility and high earnings dispersion; these controls do not explain away the recession coefficient.

Interpretation. The business-cycle state matters over and above simple “high uncertainty” proxies used separately.

4.7 Career concerns and herding

If managers herd more in recessions due to career concerns, one could mistakenly interpret recession timing as skill. The authors therefore test whether portfolio dispersion falls in recessions. Instead, they find the opposite: **dispersion rises**. Younger managers do herd more on average, but herding is *weaker* in recessions, not stronger.

Interpretation. The timing result is not easily explained as recession herding.

4.8 What the paper can and cannot distinguish

- The paper is strong on showing state-contingent task performance.
- It is weaker on whether managers use private information, public information, or superior processing of public data.
- The paper also does not claim that the average active fund outperforms. Rather, a **subset** of managers appears to generate value.
- It is not a fully causal macro shock paper. The main claim is that conditioning on the state of the economy reveals skill that is invisible in unconditional tests.

4.9 Relation to the literature

The paper sits at the intersection of three literatures:

- mutual fund performance and stock-picking,
- market timing,
- and business-cycle variation in managerial behavior.

Its main advance is to unify picking and timing as different uses of the same underlying cognitive skill.

5 Tables and figures

5.1 Table I: baseline cyclical skill

Read:

- The recession coefficient in the Timing regression is about **+0.140** per month.
- The recession coefficient in the Picking regression is about **-0.144** per month.
- Interpreted in annualized terms, timing is roughly **1.67% per year higher** in recessions and picking is roughly **1.75% per year lower** in recessions.

Economic message. This is the paper's central fact: average managers look like stock pickers in booms and market timers in recessions.

Table I
Timing and Picking Skills Are Cyclical

The dependent variables are *Timing* and *Picking*, defined in equations (1) and (2), where each stock's beta is measured over a 12-month rolling window. *Recession* is an indicator variable equal to one for every month the economy is in a recession according to the NBER, and zero otherwise. *Log(Age)* is the natural logarithm of fund age in years. *Log(TNA)* is the natural logarithm of fund total net assets. *Expenses* is the fund expense ratio. *Turnover* is the fund turnover ratio. *Flow* is the percentage growth in a fund's new money. *Load* is the total fund load. The last three control variables measure the style of a fund along the *Size*, *Value*, and *Momentum* dimensions, based on the average scores of stocks in the fund's portfolio in that month sorted into quintiles along each respective characteristic. All control variables are demeaned. *Flow* and *Turnover* are winsorized at the 1% level. The data are monthly and cover the period 1980 to 2005. Standard errors (in parentheses) are clustered by fund and time.

	Timing		Picking	
	(1)	(2)	(3)	(4)
Recession	0.140 (0.070)	0.139 (0.068)	-0.144 (0.047)	-0.146 (0.047)
Log(Age)		0.006 (0.006)		0.004 (0.004)
Log(TNA)		0.000 (0.003)		-0.003 (0.003)
Expenses		0.677 (1.150)		-0.636 (0.537)
Turnover		0.008 (0.011)		0.012 (0.007)
Flow		0.003 (0.077)		0.044 (0.078)
Load		0.066 (0.178)		0.142 (0.106)
Size		-0.009 (0.009)		0.005 (0.007)
Value		-0.015 (0.013)		0.030 (0.010)
Momentum		-0.015 (0.034)		0.034 (0.034)
Constant	0.907 (0.024)	0.907 (0.024)	-0.910 (0.018)	-0.910 (0.018)
Observations	221,306	221,306	221,306	221,306

Table II
Timing and Picking with Real-Time Recession Indicators

The dependent variables are *Timing* and *Picking*. *RecRT* is a recession measure based on the Chauvet and Piger (2008) real-time recession probability. *RecRT* is a continuous variable and is expressed in %; its mean is 7.52 and its standard deviation is 17. *RecCFNAI* is the Chicago Fed National Activity Index, multiplied by -1. *RecCFNAI* is a continuous variable and has mean of 0.08 and standard deviation of 0.54. All other controls are defined in Table I. All independent variables, including *RecRT* and *RecCFNAI*, are demeaned in the regression. The data are monthly and cover the period 1980 to 2005. Standard errors (in parentheses) are clustered by fund and time.

	Timing		Picking	
	(1)	(2)	(3)	(4)
RecRT	0.004 (0.002)		-0.002 (0.001)	
RecCFNAI		0.094 (0.058)		-0.059 (0.029)
Log(Age)	0.008 (0.007)	0.007 (0.007)	0.004 (0.004)	0.004 (0.004)
Log(TNA)	-0.001 (0.003)	-0.001 (0.003)	-0.003 (0.003)	-0.003 (0.003)
Expenses	0.552 (1.186)	0.524 (1.069)	-0.607 (0.543)	-0.358 (0.550)
Turnover	0.010 (0.011)	0.009 (0.010)	0.012 (0.007)	0.012 (0.007)
Flow	-0.002 (0.078)	-0.002 (0.077)	0.042 (0.078)	0.051 (0.079)
Load	0.092 (0.188)	0.093 (0.176)	0.136 (0.108)	0.111 (0.108)
Size	-0.008 (0.008)	-0.009 (0.009)	0.006 (0.007)	0.005 (0.007)
Value	-0.019 (0.013)	-0.017 (0.012)	0.029 (0.010)	0.028 (0.010)
Momentum	-0.019 (0.035)	-0.018 (0.036)	0.036 (0.034)	0.032 (0.033)
Constant	0.019 (0.024)	0.019 (0.024)	-0.022 (0.017)	-0.022 (0.017)
Observations	221,292	221,292	221,292	221,292

5.2 Table II: real-time recession indicators

Read:

- Higher real-time recession probability (RecRT) raises Timing and lowers Picking.
- A more recessionary CFNAI reading does the same.
- The signs and magnitudes are close to those in the NBER-based regressions.

Economic message. The main result is not an artifact of looking backward with perfect recession dating.

5.3 Table III: cyclical variation is strongest for the best managers

Read:

- At the median, the recession effect on Timing is about **0.059**.
- At the 95th percentile, the recession effect on Timing rises to about **0.251**.
- For Picking, the recession effect becomes more negative moving from the median to the upper tail.

Economic message. The managers who appear most skilled are also the ones who change tasks the most over the cycle.

Table III
Cyclical Variation in Timing and Picking in the Cross-Section of Funds

The dependent variables are *Timing* and *Picking*, defined in equations (1) and (2). The independent variables are the same as in Table I. This table shows results from estimating quantile regression models at the median (columns (1) and (4), Q50), 75th percentile (columns (2) and (5), Q75), and 95th percentile (columns (3) and (6), Q95) of the cross-sectional distribution (across funds) of *Timing* and *Picking*. Standard errors are computed using block bootstrapping, where the block is a cluster of analysis as in Luetkepohl (1993).

	Timing			Picking		
	Q50 (1)	Q75 (2)	Q95 (3)	Q50 (4)	Q75 (5)	Q95 (6)
Recession	0.059 (0.023)	0.114 (0.041)	0.251 (0.082)	-0.084 (0.021)	-0.091 (0.022)	-0.173 (0.067)
Log(Age)	0.000 (0.001)	-0.003 (0.004)	-0.020 (0.017)	0.003 (0.002)	-0.005 (0.003)	-0.057 (0.010)
Log(TNA)	0.000 (0.001)	0.004 (0.003)	-0.004 (0.010)	-0.001 (0.001)	0.001 (0.002)	0.005 (0.007)
Expenses	0.162 (0.258)	4.015 (1.036)	21.046 (3.464)	-0.588 (0.277)	3.096 (0.597)	18.869 (1.842)
Turnover	0.001 (0.001)	0.053 (0.012)	0.404 (0.042)	0.001 (0.001)	0.042 (0.006)	0.305 (0.031)
Flow	0.004 (0.011)	0.036 (0.048)	0.228 (0.218)	0.035 (0.021)	0.099 (0.041)	0.192 (0.140)
Load	-0.013 (0.028)	-0.327 (0.110)	-1.404 (0.475)	0.108 (0.036)	-0.129 (0.078)	-1.213 (0.248)
Size	0.000 (0.001)	-0.015 (0.005)	-0.071 (0.026)	0.005 (0.003)	-0.026 (0.005)	-0.130 (0.019)
Value	-0.001 (0.002)	-0.031 (0.010)	-0.172 (0.044)	0.015 (0.004)	-0.006 (0.006)	-0.046 (0.021)
Momentum	-0.001 (0.005)	0.037 (0.019)	0.196 (0.072)	0.013 (0.009)	0.071 (0.013)	0.278 (0.047)
Constant	0.000 (0.004)	0.108 (0.020)	0.765 (0.061)	-0.015 (0.005)	0.126 (0.013)	0.722 (0.053)
Observations	221,306	221,306	221,306	221,306	221,306	221,306

Table IV
The Same Funds Switch Strategies

We divide all fund-month observations into recession and expansion subsamples. *Expansion* = $1 - \text{Recession}$. *Top* is an indicator variable equal to one for all funds whose *Picking* in expansion is in the highest 25th percentile of the distribution, and zero otherwise. Control variables, sample period, and standard errors are described in Table I.

	Timing		Picking	
	Expansion (1)	Recession (2)	Expansion (3)	Recession (4)
Top	-0.001 (0.004)	0.037 (0.013)	0.059 (0.005)	-0.054 (0.017)
Log(Age)	0.009 (0.002)	-0.015 (0.006)	-0.001 (0.002)	0.027 (0.007)
Log(TNA)	-0.001 (0.001)	0.004 (0.003)	-0.001 (0.001)	-0.024 (0.003)
Expenses	0.571 (0.322)	0.981 (1.085)	-0.985 (0.366)	-3.491 (1.355)
Turnover	0.010 (0.003)	0.009 (0.008)	0.013 (0.004)	-0.005 (0.012)
Flow	0.058 (0.024)	-0.852 (0.112)	0.127 (0.036)	-0.054 (0.092)
Load	0.124 (0.050)	0.156 (0.162)	0.104 (0.054)	0.504 (0.197)
Size	-0.009 (0.002)	-0.057 (0.006)	0.011 (0.002)	0.023 (0.007)
Value	-0.018 (0.003)	-0.057 (0.010)	0.027 (0.003)	0.107 (0.011)
Momentum	-0.007 (0.003)	-0.148 (0.010)	0.031 (0.004)	-0.007 (0.011)
Constant	0.018 (0.001)	0.055 (0.005)	-0.022 (0.002)	-0.159 (0.006)
Observations	204,311	18,354	204,311	18,354

5.4 Table IV: the same funds switch strategies

Read:

- Funds that are top stock pickers in expansions have **no** special timing ability in expansions.
- The same funds have **significantly better timing in recessions** (about **0.037** per month).
- They have superior picking in expansions (about **0.059** per month) but worse picking in recessions.

Economic message. This is the cleanest evidence that the same managers are not permanently “good at one thing.” They switch the way they generate value.

5.5 Table V: manager-level evidence

Read:

- Using managers rather than funds as the unit of observation yields the same sign pattern.
- With manager fixed effects, the recession coefficients remain about **+0.160** for Timing and **-0.187** for Picking.

Economic message. The result is about managers, not just about organizational attributes of

funds.

Table V
Managers as the Unit of Observation

The dependent variables are *Timing* and *Picking*, defined in equations (1) and (2), both tracked at the manager level. In columns (3) and (4), we include manager fixed effects. Control variables, sample period, and standard errors are described in Table I.

	Timing		Picking	
	(1)	(2)	(3)	(4)
Recession	0.156 (0.074)	-0.192 (0.053)	0.160 (0.074)	-0.187 (0.054)
Log(Age)	0.005 (0.005)	0.002 (0.004)	0.004 (0.009)	-0.003 (0.007)
Log(TNA)	0.001 (0.003)	-0.001 (0.004)	0.004 (0.005)	0.002 (0.008)
Expenses	0.194 (1.205)	-0.586 (0.596)	0.569 (1.206)	-0.491 (0.803)
Turnover	0.008 (0.013)	0.012 (0.008)	0.004 (0.010)	0.011 (0.008)
Flow	0.014 (0.094)	0.181 (0.177)	0.019 (0.095)	0.166 (0.184)
Load	0.170 (0.209)	0.051 (0.121)	0.223 (0.191)	-0.016 (0.185)
Size	-0.009 (0.010)	0.007 (0.008)	-0.015 (0.019)	0.005 (0.011)
Value	-0.029 (0.016)	0.030 (0.010)	-0.033 (0.019)	0.021 (0.014)
Momentum	-0.022 (0.039)	0.036 (0.042)	-0.026 (0.048)	0.037 (0.048)
Constant	0.012 (0.026)	-0.009 (0.021)	0.012 (0.027)	-0.010 (0.022)
Manager	N	N	Y	Y
Fixed Effect				
Observations	333,582	333,582	333,582	333,582

Table VI
Strategy Switchers Outperform

The dependent variables—CAPM alpha, three-factor alpha, and four-factor alpha—are obtained from a 12-month rolling-window regression of a fund's excess returns, before expenses, on a set of common risk factors. *Top* is an indicator variable equal to one for all funds whose *Picking* in expansion is in the highest 25th percentile of the distribution, and zero otherwise. Control variables, sample period, and standard errors are described in Table I. *Expansion* equals one every month the economy is not in recession according to the NBER, and zero otherwise.

	CAPM alpha	Three-Factor Alpha	Four-Factor Alpha
	(1)	(2)	(3)
Top	0.068 (0.028)	0.040 (0.018)	0.058 (0.016)
Log(Age)	-0.035 (0.008)	-0.029 (0.006)	-0.038 (0.006)
Log(TNA)	0.036 (0.005)	0.013 (0.004)	0.014 (0.003)
Expenses	4.972 (0.942)	0.187 (0.777)	0.070 (0.716)
Turnover	-0.002 (0.012)	-0.049 (0.011)	-0.042 (0.008)
Flow	2.543 (0.181)	1.765 (0.101)	1.608 (0.101)
Load	-0.649 (0.186)	-0.067 (0.138)	-0.282 (0.146)
Size	-0.057 (0.025)	0.001 (0.008)	-0.000 (0.009)
Value	0.125 (0.044)	-0.061 (0.025)	-0.020 (0.017)
Momentum	0.296 (0.038)	0.184 (0.028)	0.177 (0.023)
Constant	0.058 (0.020)	0.041 (0.016)	0.050 (0.019)
Observations	226,769	226,769	226,769

5.6 Table VI: strategy switchers outperform

Read:

- Strategy-switching top funds earn around **0.040 to 0.068** monthly alpha depending on whether performance is measured with CAPM, three-factor, or four-factor models.
- These translate into roughly **48 to 82 basis points per year** of extra alpha.

Economic message. The switching behavior is not just statistically interesting. It is associated with economically meaningful outperformance.

5.7 Table VII: who the skilled funds are and how they time the market

Read:

- Top funds are **younger, smaller, more expensive**, and have much **higher turnover**.
- They receive stronger inflows and are more likely to be managed by people who later depart to hedge funds.
- In recessions, funds increase cash holdings, reduce equity beta, and rotate toward defensive sectors such as healthcare, nondurables, wholesale, and utilities, while cutting exposure to cyclical sectors such as telecom, business equipment, manufacturing, energy, and durables.

Economic message. Timing is implemented through concrete portfolio choices, not through an abstract latent factor alone.

5.8 Tables VIII and IX: the Skill Index predicts returns

Read:

- Skill-quintile portfolio sorts show a monotonic relation between skill and future performance.

Table VII
Comparing Top Funds to Other Funds

We divide all fund-month observations into recession and expansion subsamples. Expansion equals one every month the economy is not in recession according to the NBER, and zero otherwise. *Top* is one for any fund with *Picking* (defined in Table 1) in the highest 25th percentile in expansion, and zero otherwise. Panel A reports fund-level characteristics. *Age*, *TNA*, *Expenses*, *Turnover*, and *Flow* are defined in Table 1. *RSI* comes from Kuperovsk, Salas, and Zheng (2008). *Portfolio Dispersion* is the concentration of the fund's portfolio, measured as the Herfindahl index of portfolio weights in deviation from the market portfolio's weights. *Stock Number* is the number of stocks in the fund's portfolio. *Industry* is the industry concentration of the fund's portfolio, measured as the Herfindahl index of portfolio weights in a given industry in deviation from the market portfolio's weights. *Beta Deviation* is the absolute difference between the fund's beta and the average beta in its style category. Panel B reports manager-level characteristics. *MBA* or *Ivy* equals one if the manager obtained an MBA degree or graduated from an Ivy League institution, and zero otherwise. *Age* and *Experience* are the fund manager's age and experience in years. *Gender* equals one if the manager is a male and zero if female. *Hedge Fund* equals one if the manager departed to a hedge fund, and zero otherwise. *Top1-Top0* is the difference between the mean values of the groups for which *Top* equals one and zero, respectively. *p-values* measure statistical significance of the difference. The data are monthly from 1980 to 2005.

	Top = 1			Top = 0			Difference	
	Mean	Stddev	Median	Mean	Stddev	Median	Top1-Top0	p-value
Panel A: Fund Characteristics								
Age	10.01	8.91	7	15.20	15.34	9	-5.19	0.000
TNA	621.13	2,037.04	129.00	1,019.45	4,024.29	162.90	-398.32	0.002
Expenses	1.40	0.47	1.42	1.22	0.47	1.17	0.26	0.000
Turnover	130.41	166.44	101.00	79.89	116.02	58.00	50.52	0.000
Flow	0.22	7.39	-0.76	-0.07	6.47	-0.73	0.300	0.008
Portfolio Dispersion	1.68	1.60	1.29	1.33	1.50	0.99	0.35	0.000
Stock Number	90.83	110.20	68	111.86	187.13	69	-21.03	0.000
Industry	6.40	7.90	6.39	5.97	7.54	3.54	0.43	0.000
Beta Deviation	0.18	0.38	0.13	0.13	0.23	0.10	0.05	0.000
RSI	4.13	5.93	1.82	2.77	3.97	1.26	1.37	0.000
Panel B: Fund Manager Characteristics								
MBA	42.09	49.27	0	39.49	48.88	0	2.60	0.128
Ivy	35.36	43.21	0	27.94	44.87	0	7.47	0.205
Age	33.02	10.42	30	54.11	10.06	52	-1.08	0.081
Experience	26.45	10.01	24	28.14	10.00	26	-1.69	0.003
Gender	90.89	28.77	100	90.50	29.31	100	0.39	0.681
Hedge Fund	10.43	30.57	0	6.12	23.96	0	4.31	0.000

Table VIII
Skill Index Portfolio Sorts

Each month we sort mutual funds into five quintiles based on their *Skill Index*, defined in equation (7), from lowest values (Q1) to highest values (Q5). We report equal-weighted average abnormal returns (Panel A), CAPM alpha (Panel B), Fama-French three-factor alpha (Panel C), and Fama-French-Carhart four-factor alpha (Panel D) of each quintile portfolio 1, 5, 6, 9, or 12 months after portfolio formation. The last row of each table (Average) reports the average abnormal return or alpha across all funds. All numbers represent monthly returns (in %).

Panel A: Abnormal Returns					Panel C: FF 3F Alpha					
1 mo	3 mo	6 mo	9 mo	12 mo	1 mo	3 mo	6 mo	9 mo	12 mo	
Q1	-0.282	-0.076	-0.240	-0.049	-0.017	Q1	-0.176	-0.180	-0.193	-0.198
Q2	-0.121	-0.059	-0.109	-0.094	-0.041	Q2	-0.101	-0.101	-0.104	-0.110
Q3	-0.052	-0.047	-0.065	-0.094	-0.064	Q3	-0.067	-0.065	-0.062	-0.065
Q4	0.033	-0.015	0.074	-0.086	-0.032	Q4	-0.019	-0.018	-0.020	-0.018
Q5	0.250	0.037	0.263	0.081	-0.007	Q5	0.117	0.108	0.101	0.095
Average	-0.030	-0.032	-0.032	-0.032	-0.032	Average	-0.049	-0.051	-0.055	-0.059
Panel B: CAPM Alpha					Panel D: Carhart 4F Alpha					
1 mo	3 mo	6 mo	9 mo	12 mo	1 mo	3 mo	6 mo	9 mo	12 mo	
Q1	-0.271	-0.289	-0.283	-0.304	-0.107	Q1	-0.144	-0.146	-0.152	-0.164
Q2	-0.138	-0.146	-0.146	-0.161	-0.086	Q2	-0.083	-0.085	-0.086	-0.086
Q3	-0.052	-0.050	-0.054	-0.061	-0.061	Q3	-0.055	-0.054	-0.051	-0.054
Q4	0.064	0.071	0.063	0.066	-0.014	Q4	-0.010	-0.009	-0.014	-0.015
Q5	0.249	0.253	0.242	0.262	0.095	Q5	0.113	0.106	0.096	0.091
Average	-0.029	-0.032	-0.035	-0.039	-0.041	Average	-0.036	-0.037	-0.041	-0.045

Table IX
Skill Index Predicts Performance

The dependent variables are, respectively, the fund's cumulative CAPM, three-factor, or four-factor alpha, calculated from a 12-month rolling-window regression. The regression window is $t - 10$ to $t + 2$ for one month ahead and $t - 1$ to $t + 15$ for one year ahead. For each fund, we form *Skill Index* defined in equation (7). *Picking* and *Timing* are defined in Table 1, except that they are normalized so that they are mean zero and have a standard deviation of one over the full sample. The other control variables, sample period, and standard error calculation are the same as in Table 1.

	One Month Ahead			One Year Ahead		
	CAPM Alpha (1)	Three-Factor Alpha (2)	Four-Factor Alpha (3)	CAPM Alpha (4)	Three-Factor Alpha (5)	Four-Factor Alpha (6)
Skill Index	0.202 (0.0380)	0.103 (0.0109)	0.094 (0.017)	0.197 (0.0280)	0.090 (0.0220)	0.091 (0.013)
Log(Age)	-0.027 (0.007)	-0.022 (0.005)	-0.033 (0.006)	-0.014 (0.007)	-0.008 (0.005)	-0.023 (0.006)
Log(TNA)	0.025 (0.004)	0.005 (0.003)	0.006 (0.003)	-0.012 (0.004)	-0.018 (0.003)	-0.012 (0.003)
Expenses	-0.347 (1.026)	-0.139 (0.797)	-0.040 (0.755)	-0.571 (0.983)	-0.423 (0.748)	-0.475 (0.660)
Turnover	-0.041 (0.010)	-0.075 (0.008)	-0.065 (0.008)	-0.007 (0.012)	-0.050 (0.011)	-0.048 (0.009)
Flow	2.226 (0.156)	1.585 (0.149)	1.430 (0.144)	0.106 (0.184)	0.163 (0.164)	0.164 (0.171)
Load	-0.655 (0.189)	-0.037 (0.154)	-0.271 (0.174)	-0.576 (0.122)	0.250 (0.132)	-0.009 (0.132)
Size	-0.011 (0.024)	0.016 (0.008)	0.012 (0.008)	-0.061 (0.028)	-0.065 (0.010)	-0.065 (0.010)
Value	0.227 (0.030)	0.010 (0.019)	0.045 (0.017)	0.235 (0.036)	0.034 (0.025)	0.074 (0.021)
Momentum	0.246 (0.042)	0.158 (0.031)	0.107 (0.025)	0.088 (0.031)	0.056 (0.030)	0.088 (0.025)
Constant	-0.032 (0.033)	-0.056 (0.031)	-0.042 (0.030)	-0.044 (0.044)	-0.071 (0.040)	-0.058 (0.021)
Observations	219,321	219,321	219,321	187,659	187,659	187,659

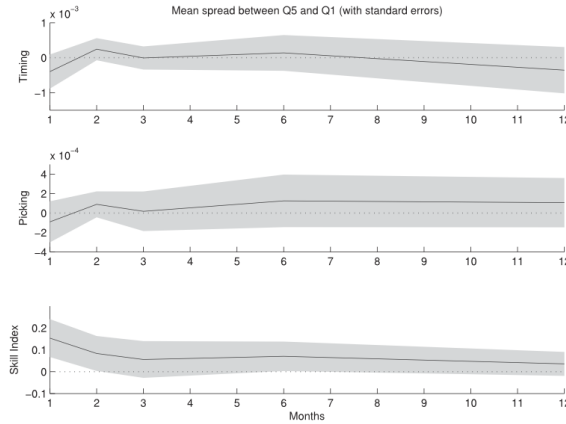


Figure 1. Persistence of Timing, Picking, and Skill Index. We rank funds into quintiles based on their *Timing*, *Picking*, or *Skill Index* score at time 0. Next, we subtract the average score in quintile 5 (Q5) from that in quintile 1 (Q1) in each of the following 12 months. We report that difference in the postformation period. A positive difference indicates persistent skill. The shading shows two standard errors on either side of the point estimate (solid line).

- The top-minus-bottom spread is roughly **3% to 6% annualized** at short horizons and still positive at longer horizons.
- In panel predictive regressions, a one-standard-deviation increase in Skill Index predicts about **2.4%** higher annualized CAPM alpha one month ahead and about **1.1% to 1.2%** higher annualized multifactor alpha.
- At the one-year horizon, the corresponding predictive effects are about **2.2%** for CAPM alpha and about **1.0%** for three- and four-factor alpha.

Economic message. A simple, real-time, state-contingent skill measure has substantial predictive content for future fund performance.

5.9 Figure 1: persistence of timing, picking, and the Skill Index

Read:

- Past Timing alone is not persistent.
- Past Picking alone is not persistent.

- The Skill Index *is* persistent: the gap between top- and bottom-skill funds remains positive for many months and statistically meaningful for roughly half a year.

Economic message. Timing and picking look fleeting on their own because managers switch tasks. Once the two are combined in a state-contingent way, the underlying skill becomes visible.

6 Short summary

- The paper asks whether mutual fund skill is state dependent.
- It measures skill directly from portfolio holdings using timing and picking formulas.
- Market timing rises in recessions, while stock picking is stronger in booms.
- The right tail of managers shows the strongest cyclical switching.
- The same top funds pick stocks well in expansions and time the market well in recessions.
- These switchers earn higher unconditional alpha.
- A real-time Skill Index that weights timing more in recessions and picking more in booms predicts future net alpha.
- The combined skill measure is persistent even though timing and picking separately are not.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that mutual fund skill is *time varying in nature*. Skilled managers appear to reallocate attention across tasks: they pick stocks in expansions and time the market in recessions. This is not just an average-fund pattern; it is strongest for top-performing funds and survives manager-level tests.

7.2 Economic intuition

When macro conditions worsen, aggregate shocks matter more for portfolio performance, so it becomes more valuable to process macro information and adjust systematic exposure. When conditions are good, aggregate risk matters less and firm-specific information becomes relatively more valuable. A manager with general information-processing ability therefore changes what that ability is used for.

7.3 Takeaway

This paper is important because it reframes the mutual fund skill debate. Instead of asking whether managers are always stock pickers or always market timers, it asks whether skilled managers know *which task matters most right now*. That perspective helps reconcile why unconditional evidence on timing often looks weak even when some managers truly add value.